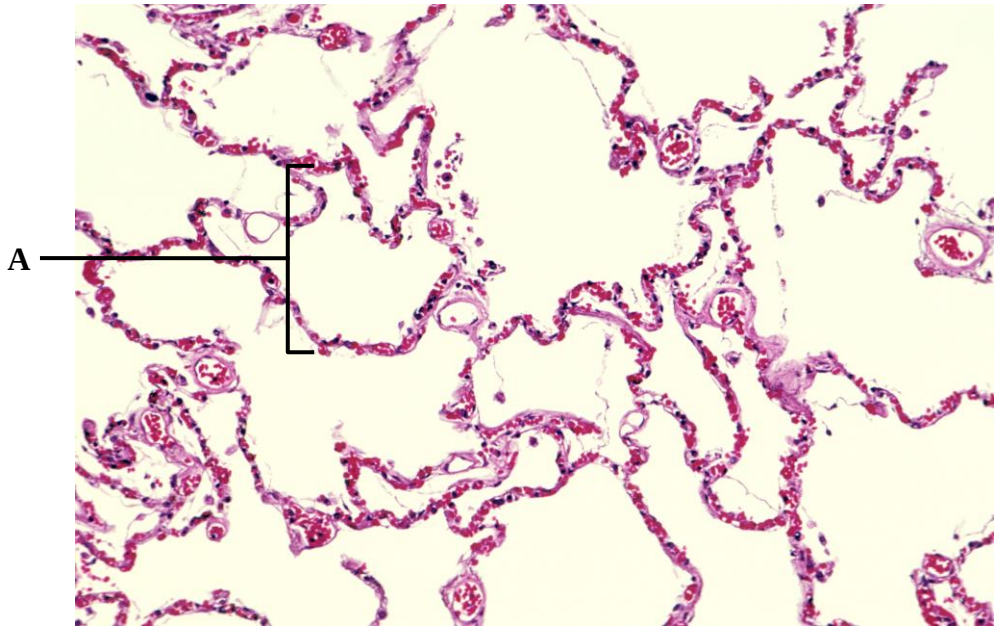
Structured questions

|!|BCELB0080000001|!| ★

The photomicrograph below shows a section of human lung tissue.



Section of human lung tissue (×180)

(a) With reference to ***two*** observable features shown in the photomicrograph, explain how the lungs are adapted for gas exchange. (4 marks)

(b) Describe how oxygen in structure A is transported to other parts of the body. (4 marks)

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(c) Emphysema (肺氣腫) is a lung condition that causes shortness of breath. This symptom is caused by the breakdown of the walls of structure A. Suggest why the breakdown of the walls of structure A causes shortness of breath. (2 marks)

_____

_____

_____

##

(a) Any ***two*** of the following: (max. 4)

- The wall of structure A is very thin (one-cell-thick). (1)
  This reduces the diffusion distance of gases. (1)
- Structure A is richly supplied with blood capillaries. (1)
  This allows a rapid transport of gases into and away from structure A, and hence maintains a steep concentration gradient for diffusion. (1)
- There is a large number of structure A in the lungs. (1)
  This provides a large surface area for gas exchange. (1)

(Or other reasonable answers)

(b) Oxygen dissolves in the water film on the inner surface of structure A. (1)
Oxygen diffuses across the walls of structure A and capillaries into the blood. (1)
It then diffuses into red blood cells and combines with haemoglobin (to form oxyhaemoglobin). The blood becomes oxygenated. (1)
Oxygenated blood returns to the heart and is pumped to other parts of the body. (1)

(c) The total surface area for gas exchange decreases, (1)
leading to a lower efficiency of gas exchange. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	2
(b)	Understanding basic concepts	4
(c)	Applying concepts	2

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|!|BCELB0080000002|!| ★

The table below shows the percentage composition of inhaled and exhaled air.

Gas	Inhaled air (%)	Exhaled air (%)
Nitrogen	78.0	78.0
Oxygen	21.0	16.0
Carbon dioxide	0.04	4.0
Water vapour	Variable	Saturated

(a) Explain why the percentage of nitrogen gas in exhaled air is the same as in inhaled air. (1 mark)

(b) Describe how oxygen in the inhaled air is carried away from the lungs. (4 marks)

(c) The carbon dioxide content of exhaled air is higher than that of inhaled air.

(i) Where is the carbon dioxide produced? (1 mark)

(ii) In what form is carbon dioxide transported in blood? (1 mark)

(d) Explain briefly why the water vapour content in exhaled air is higher that of inhaled air. (2 marks)

##

(a) Nitrogen is not used or produced by body cells. (1)

(b) Oxygen in the inhaled air dissolves into the water film on the walls of air sacs. (1)
Oxygen then diffuses into red blood cells. (1)
It binds to haemoglobin, forming oxyhaemoglobin. (1)

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The oxygenated blood is carried away from the lungs through the pulmonary veins. (1)

(c) (i) Body cells / body tissues (1)

(ii) Hydrogencarbonate ions / carboxyhemoglobin (1)

(d) Exhaled air is moistened by (1)

the mucus in the air sacs / respiratory tract / nasal cavity. (1)

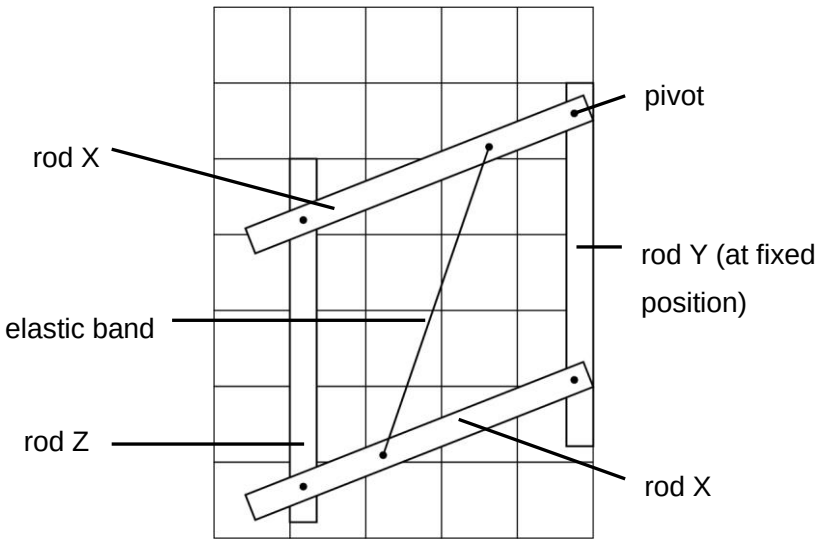
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	4
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(d)	Understanding basic concepts	2

|!|BCELB0080000003|!| ★

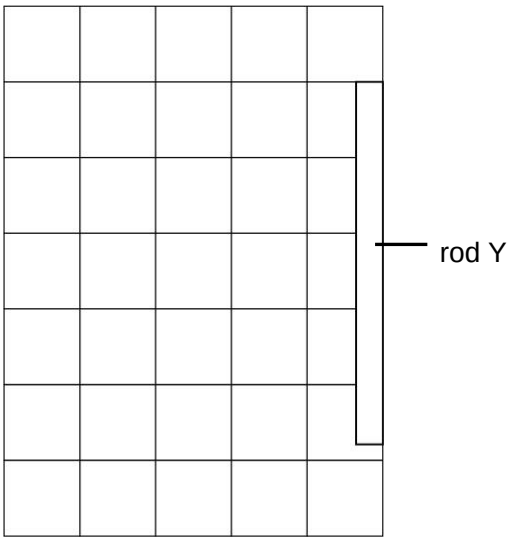
The diagram below shows a model made of rods which is used to show the position of the ribs, sternum and backbone at the end of exhalation.



(a) State the parts of the body represented by rods X, rod Y, rod Z and elastic band. (4 marks)

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(b) On the grid below, draw the position of rods X and rod Z at the end of inhalation. The position of rod Y has been drawn for you. (3 marks)



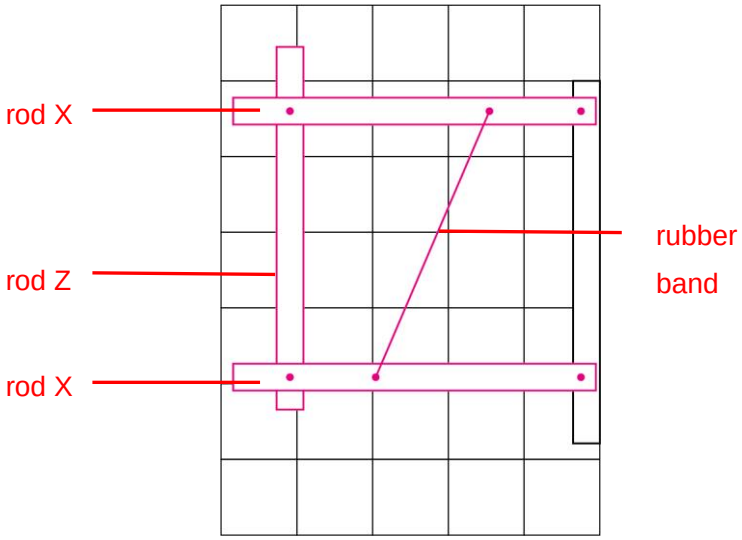
(c) Describe how the action of the diaphragm brings about inhalation. (4 marks)

##

(a) Rod X: rib (1)
Rod Y: backbone / vertebral column (1)
Rod Z: sternum (1)
Elastic band: intercostal muscle (1)

(b) Correct position of rod X (1)
Correct position of rod Z (1)
Correct position of elastic band (1)

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(c) The diaphragm muscles contract. (1)

The diaphragm becomes flattened (1)

The volume of the thoracic cavity increases, leading to a decrease in air pressure of the lungs. (1)

As the air pressure in the lungs is lower than the atmospheric pressure, air rushes into the lungs. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Presenting information	3
(c)	Understanding basic concepts	4

|!|BCELA0080000004|!| ★★

Children are at a high risk of drowning. During drowning, vital organs such as the brain and lungs cannot get enough oxygen. The victim may die within minutes. He or she can be rescued by CPR (cardiopulmonary resuscitation).

CPR involves mouth-to-mouth breathing and chest compressions. Mouth-to-mouth breathing is the act of blowing air into the victim's lungs, as shown in the diagram below.

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(a) Suggest why a victim may die within minutes during drowning. (4 marks)

(b) Explain why breathing exhaled air into the victim's lungs can help rescue the victim. (2 marks)

(c) Explain why each of the following steps is necessary during mouth-to-mouth breathing.

(i) Pinch the victim's nostrils shut. (1 mark)

(ii) Tilt the victim's head back and lift the chin. (1 mark)

(iii) Watch for the movement of the chest. (1 mark)

##

(a) The victim's respiratory tract and lungs are filled with water. (1)
Gas exchange between the air sacs and blood capillaries ceases. (1)
The blood leaving the lungs remains deoxygenated / does not contain enough oxygen. (1)

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The brain cells and cardiac muscle cannot get enough oxygen. This will cause death. (1)

(b) Any ***one*** set: (2)

- Exhaled air still contains 16% oxygen. (1)
  If it is blown into the victim's lungs, the oxygen can be supplied to the body cells for respiration. (1)
- Exhaled air contains 4% carbon dioxide. (1)
  If it is blown into the victim's lungs, the high concentration of carbon dioxide could stimulate the victim to resume breathing. (1)

(c) (i) To make sure that the exhaled air does not leak through the nostrils. (1)

(ii) To make sure that the trachea is clear from obstruction. (1)

(iii) To make sure that air reaches the victim's lungs. (1)

(9 marks)

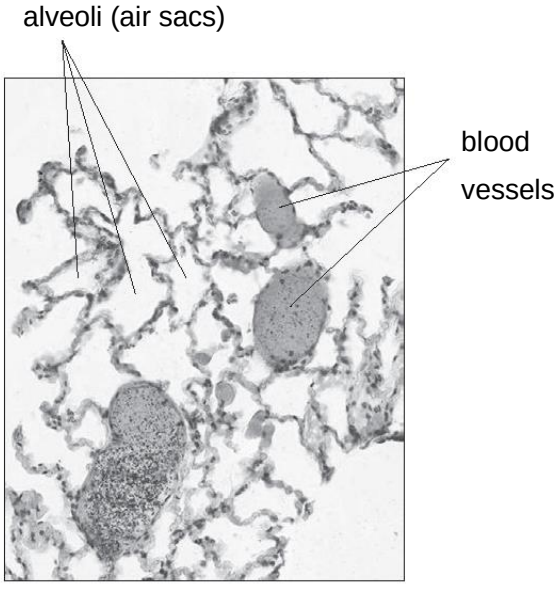
##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	1
(c)(iii)	Applying concepts	1

|!|BCELB0080000005|!| ★

WJEC GCE A/AS level Biology Module BY2 Jun 2015 Q3

Shown below is a micrograph of a section through part of a mammalian lung.



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(a) Describe and explain how ***two*** features shown in the micrograph are adaptations for efficient gas exchange. (4 marks)

(b) (i) Describe and explain the process of exhalation in a mammal. (4 marks)

(ii) Mammals have a high oxygen demand. Suggest why they need a complex ventilation mechanism (2 marks)

##

(a) Any ***two*** sets: (4)

- They have many air sacs (1)
  which increases / provides a large surface area for gas exchange. (1)
- The walls of air sacs and capillaries are thin / the capillaries are close to the air sacs (1)
  which provides a short diffusion distance for oxygen to enter the blood. (1)
- Many blood capillaries are present. (1)
  The rich blood supply maintains steep concentration gradients. (1)

(b) (i) The intercoastal muscles relax, allowing the rib cage to move downwards and inwards. (1)
The diaphragm muscles relax and the diaphragm becomes dome-shaped. (1)

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The volume of the thoracic cavity decreases, leading to an increase in air pressure of the lungs. (1)

As the air pressure in the lungs is higher than the atmospheric pressure, air moves out of the lungs. (1)

(ii) Mammals are warm-blood animals / have high metabolic rates. (1)

Ventilation maintains a steep concentration gradient at the air sacs to ensure sufficient oxygen is supplied to the air sacs. (1)

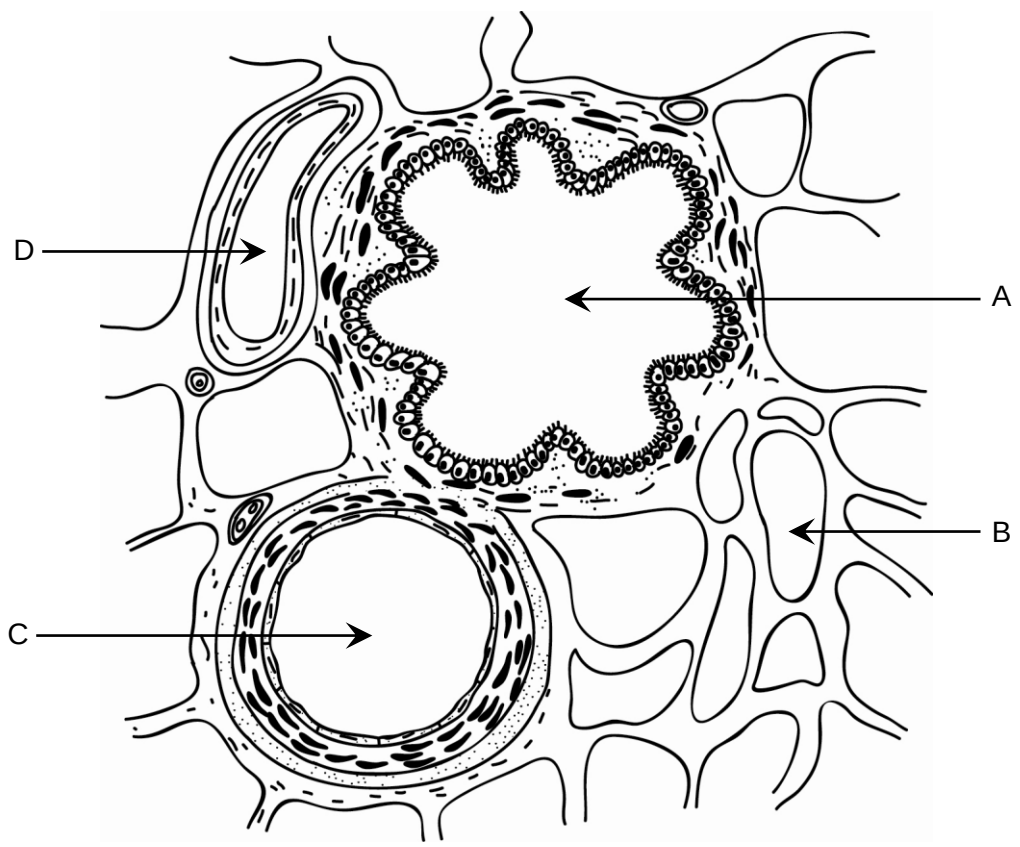
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	4
(b)(ii)	Applying concepts	2

|!|BCELA0080000006|!| ★★ **Cross topic**

The following diagram shows a section of the lung tissue.



(a) Describe how the inner wall of structure A can protect the body against bacterial infection. (6 marks)

(b) Structures C and D are blood vessels found among lung tissues. What is the difference in the blood oxygen content between structures C and D? (1 mark)

(c) The Middle East respiratory syndrome (MERS) is a viral respiratory illness and was first reported in Saudi Arabia in 2012.

(i) Suggest a possible way of transmission of MERS. (1 mark)

(ii) Structures B of MERS patients are damaged. How would the blood oxygen content of the patients be affected? (3 marks)

##

(a) Structure A is lined with mucus-secreting cells and (1)
ciliated epithelial cells. (1)
The mucus secreted by mucus-secreting cells can trap bacteria in the incoming air. (1)
The beating action of the cilia can sweep the mucus upwards to the pharynx, where the mucus is then swallowed or coughed out. (1)
The cells on the inner wall of A are closely packed, (1)
forming a physical barrier to prevent the entry of bacteria into the body. (1)

(b) Structure D has a higher oxygen content than structure C. (1)

(c) (i) Through air and droplets (1)

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(ii) The damage of structure B will decrease the surface area for dissolving oxygen, (1)

therefore reduces the rate of diffusion of dissolved oxygen into the blood capillaries. (1)

Thus the oxygen content of the blood decreases. (1)

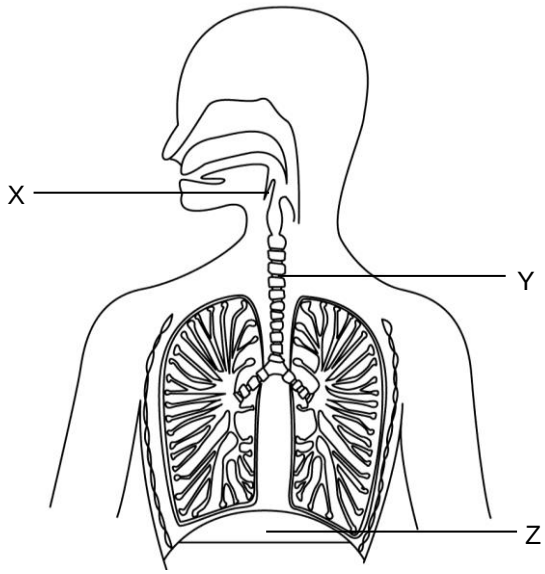
(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
	Applying concepts	3
(b)	Analysing data	1
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	3

|!|BCELC0080000007|!|

The diagram below shows a human breathing system.



(a) Name structure X and state its function. (2 marks)

(b) (i) How does a person react when food enters Y? (1 mark)

(ii) Is the reaction voluntary or involuntary? Explain you answer. (2 marks)

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(c) Describe how structure Z changes its shape during ventilation. (2 marks)

_____

_____

_____

##

(a) Epiglottis. (1)

It covers the glottis and prevents food from entering the trachea when swallowing food. (1)

(b) (i) The person would cough. (1)

(ii) Coughing is involuntary. (1)

This is because it is a reflex action and not controlled by the cerebrum. (1)

(c) During inhalation, diaphragm muscles contract and structure Z becomes flattened. (1)

During exhalation, diaphragm muscles relax and structure Z returns to its original dome shape. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCELB0080000008|!| ★

The figures below show two breathing conditions in humans.

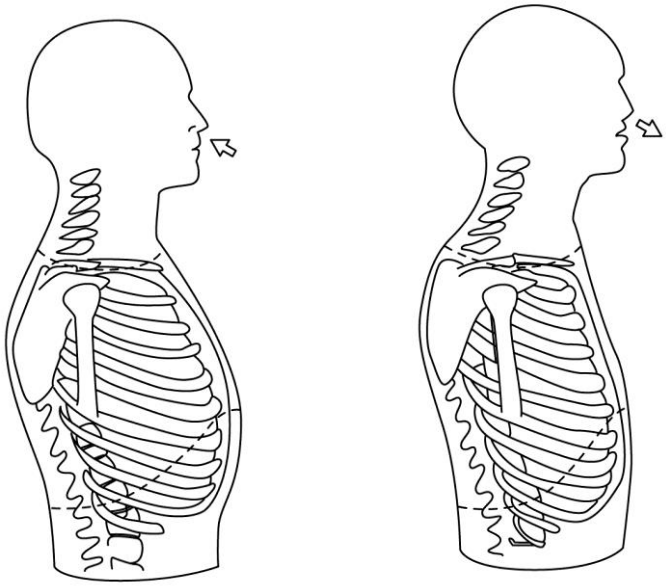


Figure A **Figure B**

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(a) Identify the phases of breathing as shown in **Figure A** and **Figure B**. (2 marks)

(b) Describe the actions that lead to the change in the volume of the thoracic cavity **from Figure A to Figure B**. (5 marks)

(c) What is the substance secreted by the pleural membranes? State its function. (2 marks)

##

(a) Figure A: inhalation (1)
Figure B: exhalation (1)
(b) The intercostal muscles relax. (1)
The ribs and the sternum move downwards and inwards. (1)
The diaphragm muscles relax, (1)
and the diaphragm returns to its original dome shape. (1)
The volume of the thoracic cavity decreases as a result. (1)
(c) Pleural fluid (1)
It reduces the friction between the lungs and the ribs during breathing. (1)

(9 marks)

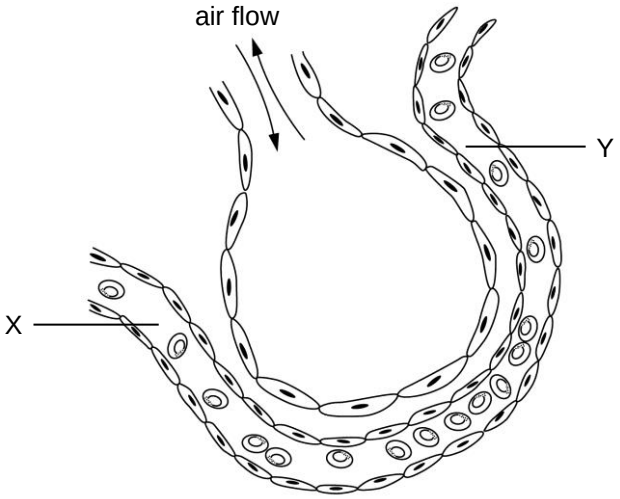
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	5
(c)	Understanding basic concepts	2

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|!|BCELA0080000009|!| ★★

The figure below shows a part of the breathing system. The part is surrounded by a capillary network.



(a) Name this part of the breathing system. (1 mark)

(b) State and explain ***two*** adaptive features of this part so that it can better carry out its function. (4 marks)

(c) (i) The air in the high mountain has less oxygen than that at the lowland. When one moves from the lowland to the high mountain, state and explain the changes in his uptake of oxygen. (3 marks)

(ii) Suggest what changes would occur in human's blood after living in the high mountain for a long period. (1 mark)

##

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(a) Air sac (1)

(b) Any ***two*** sets of the following: (4)

- The inner surface is kept moist by a water film (1)
  which allows oxygen to dissolve in it and then diffuse across the epithelium of the air sacs. (1)
- The wall of the air sacs is made up of a layer of epithelial cells (1)
  so as to provide a short distance for diffusion of gases. (1)
- There are a large number of air sacs (1)
  which provides a large surface area for gas exchange by diffusion. (1)
- The air sacs are surrounded by dense networks of blood capillaries. (1)
  This maintains a steep concentration gradient of gases between the blood and the air sacs. (1)

(c) (i) The concentration gradient of oxygen between the atmosphere and the blood is smaller, (1)
the rate of diffusion of oxygen from the inside of air sac into the bloodstream is lower. (1)
Thus, less oxygen is taken up by the body. (1)

(ii) The blood would contain more red blood cells. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	4
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	1

|!|BCELB0080000010|!| ★

Blood is essential for oxygen transport in our bodies.

(a) What is the chemical substance in blood that can help transport oxygen? (1 mark)

(b) Explain how this substance can be used as an oxygen carrier in our bodies. (2 marks)

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(c) In the past, some athletes cheated by injecting themselves with extra red blood cells before competitions. Explain how this method can improve their short-term athletic performance. (4 marks)

##

(a) Haemoglobin (1)

(b) At high oxygen concentration (in the air sacs), oxygen combines with the haemoglobin to form oxyhaemoglobin. (1)

At low oxygen concentration (in the body tissues), oxyhaemoglobin breaks down into haemoglobin and oxygen. (1)

(c) Increasing the number of red blood cells allows the blood to carry more oxygen. (1)

This leads to a higher rate of respiration (1)

so that more energy can be released for respiration. (1)

More energy can then be supplied for vigorous muscle contraction during competitions. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	4

|!|BCELB0080000011|!| ★

James and Raymond are both 22-year-old healthy men. One of them has recently been to Tibet and stayed there for three months for carrying out a research project at an altitude of 5000 m. The table below shows the blood samples taken from them.

	Number of red blood cell per litre of blood
James	4.9×10^{12}
Raymond	7.1×10^{12}

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(a) Calculate the percentage by which the number of red blood cells per litre of blood in Raymond is greater than that in James. Show your working. (2 marks)

(b) Based on the information provided, determine who has been to Tibet and explain why. (4 marks)

(c) Briefly explain how oxygen is transported from the air sacs to body cells. (4 marks)

##

(a) $(7.1 \times 10^{12} - 4.9 \times 10^{12}) \div 4.9 \times 10^{12} \times 100\%$ (1)
= 44.9% (1)

(b) Raymond (1)
His blood sample contains more red blood cells per litre of blood. (1)
At high altitude, there is less oxygen in the air. (1)
A greater number of red blood cells are needed for carrying oxygen so as to maintain the normal functioning of the body. (1)

(c) Oxygen from the air sacs of the lungs diffuses into the red blood cells (1)
and combines with haemoglobin to form oxyhaemoglobin. (1)
The oxygenated blood is transported to the body cells. (1)
Oxyhaemoglobin then breaks down to release oxygen which diffuses into body cells. (1)

(10 marks)

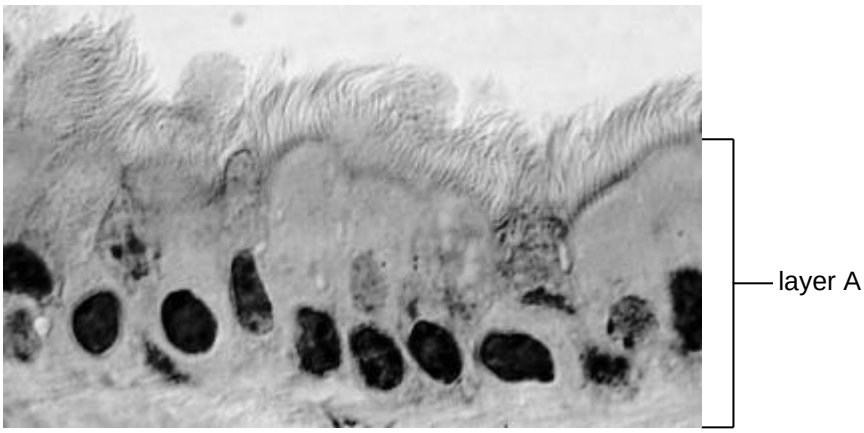
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##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	1
	Applying concepts	3
(c)	Understanding basic concepts	4

|!|BCELB0080000012|!| ★

Thomas examined the photomicrograph below that shows some cells from the lining of the respiratory tract.



(a) Name ***two*** types of cells in layer A and state their functions in keeping the respiratory tract free from dust particles. (4 marks)

Name	***Function***
(i)	
(ii)	

(b) In some Chinese cities, the residents use coal as a fuel. There would be a higher level of soot particles in the air.

(i) State the effect of the soot particles on gas exchange in lungs. (2 marks)

_____

_____

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(ii) These residents may have a higher chance in suffering from bronchitis which is the inflammation of the lining of the bronchi. Explain how bronchitis affects the breathing of the patients. (2 marks)

##

(a)

Name	***Function***	
(i) mucus-secreting cells	produce mucus to trap dust particles	(2)
(ii) ciliated epithelial cells	sweep the mucus with dust particles towards the pharynx	(2)

(b) (i) The soot particles in the air sacs reduce the surface area for gas exchange. (1)

Thus the rate of gas exchange becomes lower if more soot particles are present in the lungs. (1)

(ii) The passageways of the bronchi become narrow, (1)

making breathing more difficult. (1)

(8 marks)

##

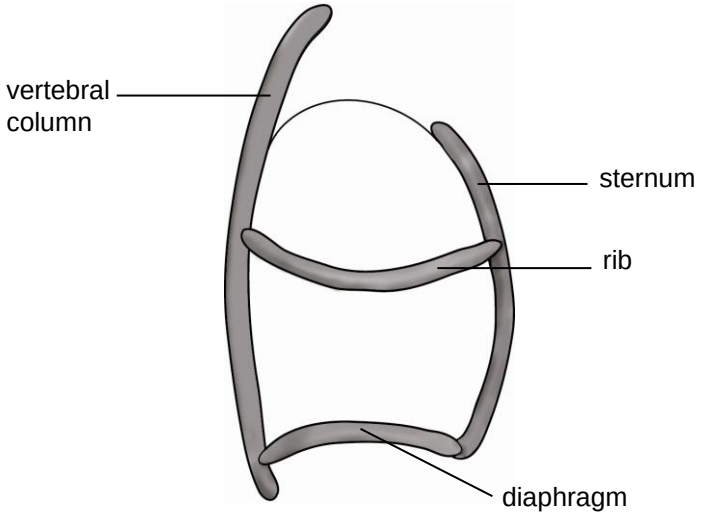
	Skills assessed	Marks
(a)(i)	Analysing data	1
	Understanding basic concepts	1
(a)(ii)	Analysing data	1
	Understanding basic concepts	1
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2

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|!|BCELC0080000013|!|

In an investigation, Vivian was asked to put her right hand on her sternum and put her left hand on the left side of her rib cage. She found that both the sternum and rib cage moved when she breathed.

(a) On the diagram shown below, draw the positions of the labelled parts after exhalation. (3 marks)



(b) There are muscles in the diaphragm. Describe the role of these muscles in exhalation. (4 marks)

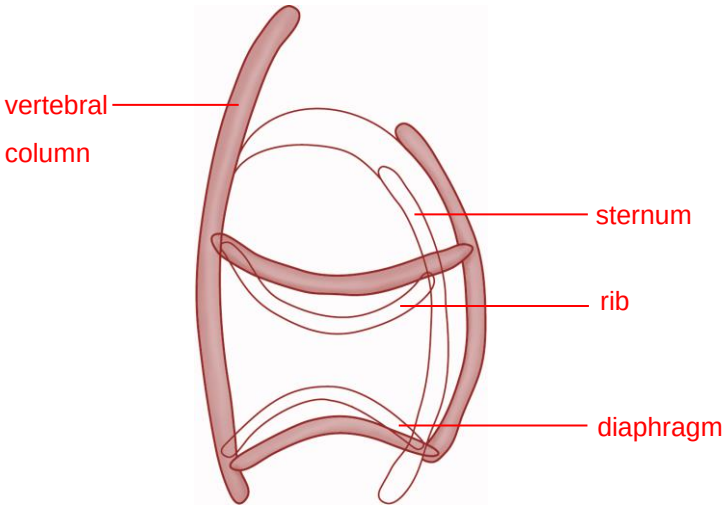
(c) During breathing, the ribs move. Name the muscles involved. (1 mark)

(d) The ribs are joined to the sternum by cartilage. What is the significance of this? (1 mark)

##

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(a)



During exhalation

Correct position of the sternum (1)
Correct position of the rib (1)
Correct position of the diaphragm (1)

(b) The diaphragm muscles relax (1)
and the diaphragm returns to its dome shape. (1)
This reduces the volume of the thoracic cavity and (1)
the air pressure in the lungs increases. (1)

(c) Intercostal muscles (1)

(d) To allow flexibility for the ribcage to move during breathing. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCELC0080000014|!|

Read the passage below and answer the questions that follow.

Larynx

Your larynx is a framework of cartilages, muscles and elastic membranes that forms the entrance to your trachea. The larynx is the part at the top of the trachea. The epiglottis is a small lid that prevents food and drinks from entering your trachea during swallowing. Therefore, you cannot swallow and breathe at the same time. Don't talk too much when you are eating. Otherwise you may choke when food wrongly enters the trachea.

(a) Explain why swallowing and breathing cannot occur at the same time. (2 marks)

(b) The pharynx is a common passageway for both food and air. Breathing through the mouth is possible, but it is not a good habit. State ***three*** advantages of breathing through the nose. (3 marks)

(c) Name the structure which supports the trachea. (1 mark)

(d) Describe how sound is produced in the larynx. (1 mark)

##

(a) The epiglottis covers the glottis when one swallows. Therefore, breathing stops temporarily. (1)

The epiglottis can block the entrance of the trachea to prevent food from entering the trachea, and thus prevents choking. (1)

(b) Nostrils are lined with hairs which can filter large dust particles from the incoming air. (1)

The wall of the nasal cavity is lined with mucus-secreting cells which secrete mucus to moisten the incoming air. (1)

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The wall of the nasal cavity is also richly supplied with blood capillaries and the blood can warm the incoming air. (1)

(c) (C-shaped rings of) cartilage (1)

(d) When air is forced through the vocal cords in the larynx, the vocal cords vibrate to produce sound. (1)

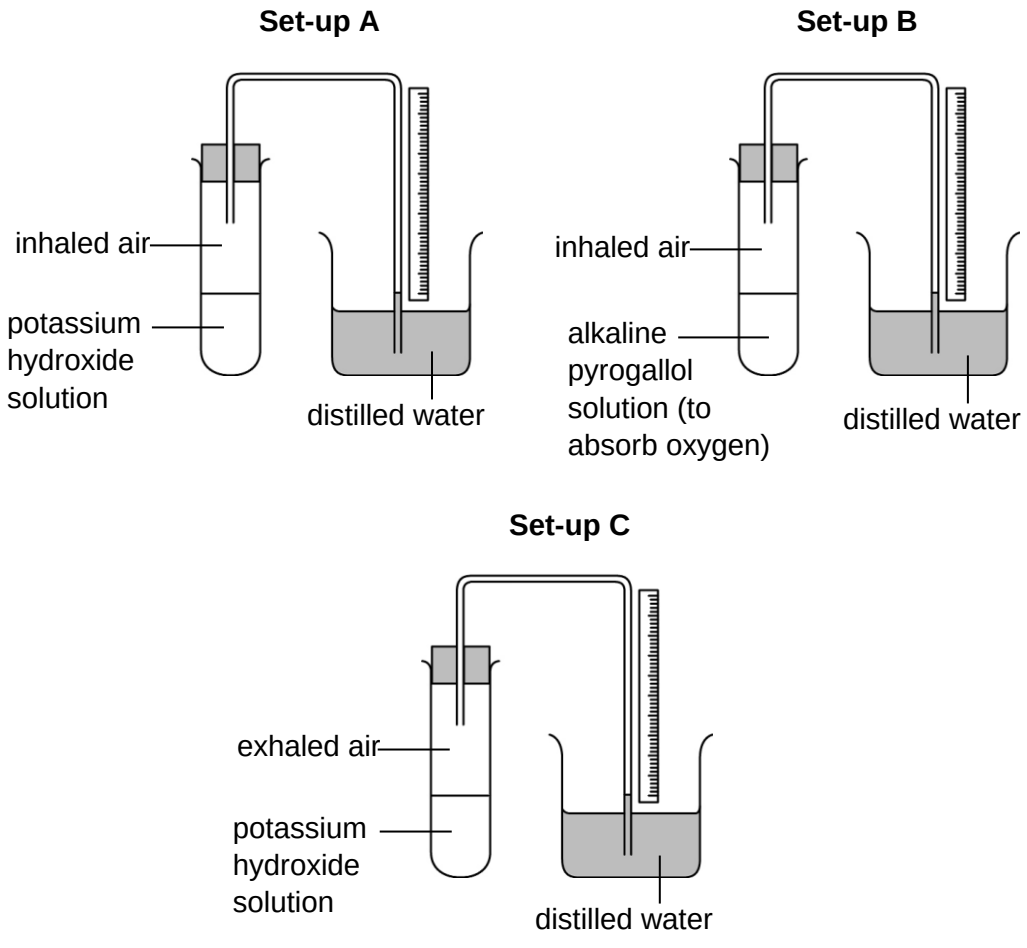
(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	3
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCELB0080000015|!| ★ SI

The following diagram shows the experimental set-ups that were used to examine the composition of air samples. In each set-up, the air sample was treated with a chemical, and the water level in the capillary was measured 30 minutes later.



The following table shows the results.

Set-up	Increase in the water level of the capillary tube (cm)
A	0
B	21
C	3.5

(a) What is the function of potassium hydroxide solution? (1 mark)

(b) Explain the results of set-up A. (3 marks)

(c) Given that the diameter of the lumen of the capillary tube is 0.3 cm, what is the change in the volume of the treated air sample in set-up B? Show your working. (2 marks)

(d) Compare the results of set-ups A and C. What conclusion can you draw from the comparison? Explain the difference. (3 marks)

##

(a) To absorb carbon dioxide (1)

(b) Since the percentage of carbon dioxide in the inhaled air is only about 0.03%, (1)
the volume of carbon dioxide being absorbed by potassium hydroxide solution in set-up A is very small (1)
this leads to no change in the water level. (1)

(c) Change in the volume of the treated air sample in set-up B
$= 0.15 \times 0.15 \times 3.14 \times 21$ (1)
$= 1.48\ cm^3$ decrease in the volume (1)

(d) The exhaled air has more carbon dioxide than inhaled air. (1)

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This is because carbon dioxide is produced by cells as a waste product of respiration, (1)

and the carbon dioxide is carried away from the body in the exhaled air. (1)

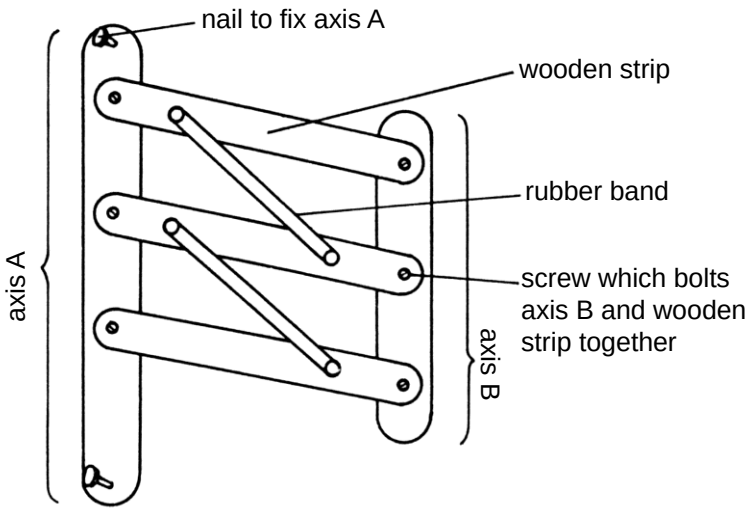
(9 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Applying concepts	3
(c)	Mathematical skills	2
(d)	Analysing data	1
	Applying concepts	2

|!|BCELB0080000016|!| ★

The diagram below shows a human rib cage model. Axis A is fixed while axis B can be moved up and down.



(a) Which parts of the body are represented by axis A, axis B, the wooden strips and the rubber bands respectively? (2 marks)

(b) What corresponding movement in the human body is demonstrated when axis B is moved up? (1 mark)

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(c) How is the movement in (b) brought about in the human body? (1 mark)

(d) What would happen to the thoracic cavity and the lungs as a result of the movement in (b)? (2 marks)

(e) State ***two*** limitations of this rib cage model in representing the actual conditions in the human body? (2 marks)

##

(a) Deduct 1 mark for one wrong answer (2)
Axis A: vertebral column
Axis B: sternum
Wooden strips: ribs
Rubber bands: intercostal muscles

(b) The sternum is moved up / Rib cage is raised. (1)

(c) by the contraction of intercostal muscles (1)

(d) The volume of thoracic cavity increases. (1)
The lungs become inflated. (1)

(e) The rubber band is shortened when its tension decreases, whereas in the human body the intercostal muscles are shortened when they contract. (1)
The tension of the rubber band is controlled by applying an external force on the rods, whereas in the human body, it is the contraction and relaxation of the intercostal muscles that bring about the movements of the ribs. (1)

(8 marks)

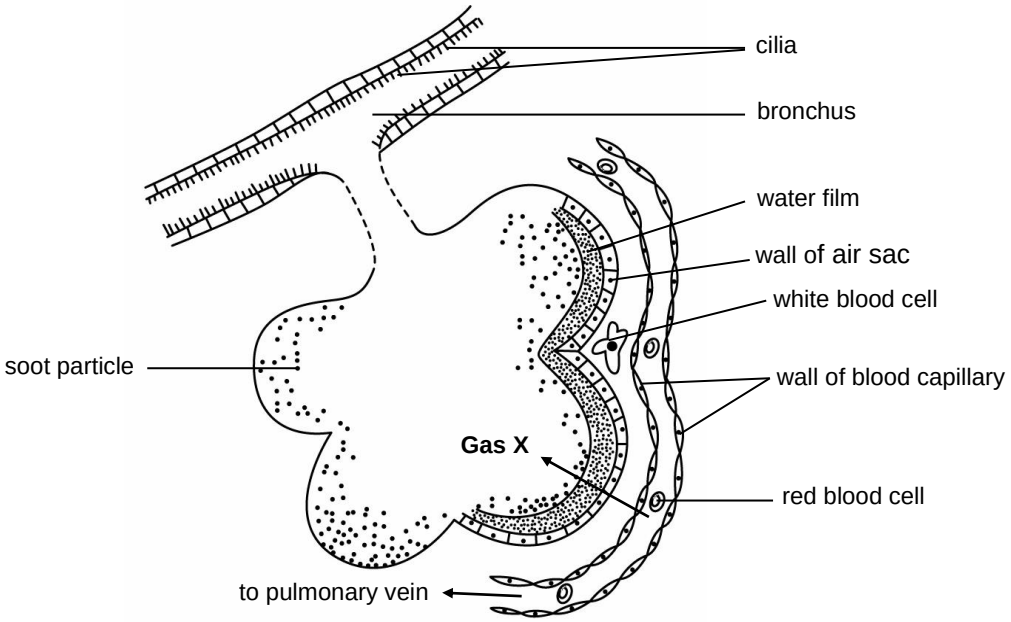
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	2

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|!|BCELC0080000017|!|

The diagram below shows the structure of the bronchus and an air sac in a human lung.



(a) State the function of the water film on the inner surface of the air sac. (1 mark)

(b) Name ***two*** gases which are the most abundant in the exhaled air. (1 mark)

(c) Describe how gas X enters the air sac from the blood capillary. (4 marks)

(d) Give ***two*** structural features of the air sacs which make the lungs efficient for gas exchange. (2 marks)

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(e) State ***one*** structure in the diagram that can protect our bodies against bacteria from the inhaled air. (1 mark)

##

(a) As a moist layer for dissolving oxygen in the air (1)

(b) Oxygen, nitrogen (1)

(c) When the blood reaches the air sacs, the hydrogencarbonate ions enter the red blood cells (1)
and break down into carbon dioxide and water. (1)
The carbon dioxide concentration in the air sac is lower than that in the blood. (1)
This causes carbon dioxide to diffuse from the blood into the air in the air sacs. (1)

(d) Any ***two*** of the following: (2)
- The wall is thin enough for rapid diffusion of gases. (1)
- The water film covering the inner surface can allow gases to dissolve in it. (1)
- It is richly supplied with blood capillaries for transport of gases / to maintain a steep concentration gradient between the blood and the air sacs. (1)

(e) Cilia for sweeping mucus which traps bacteria towards the pharynx /
White blood cells kill bacteria (1)

(9 marks)

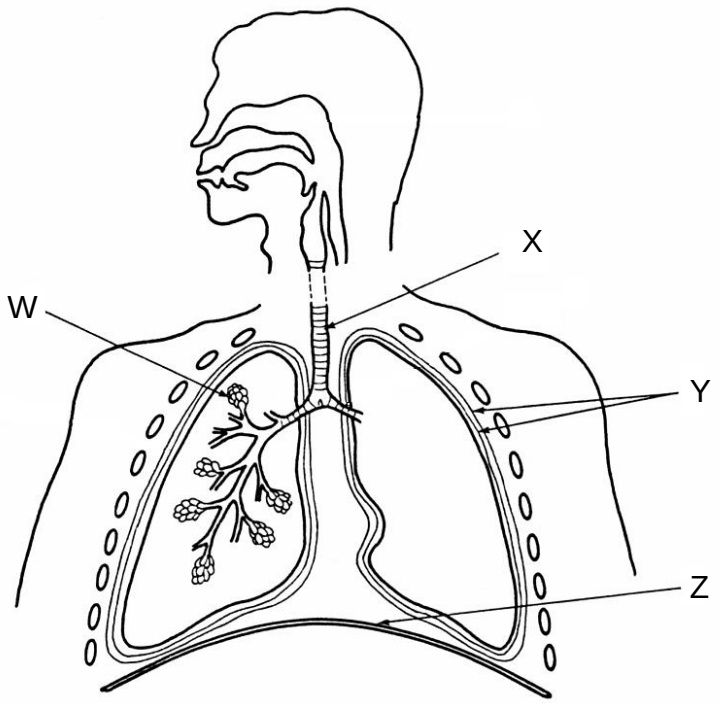
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	4
(d)	Understanding basic concepts	2
(e)	Analysing data	1

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|!|BCELB0080000018|!| ★

The following diagram shows the human breathing system.



(a) State and explain ***two*** adaptive features possessed by structure X to perform its function. (4 marks)

(b) How is structure W adapted for gas exchange? State ***two*** of the adaptations and the importance of these adaptations. (4 marks)

(c) State and explain the shape of structure Z when a man inhales. (2 marks)

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(d) State and explain what will happen if structure Y is punctured in an accident.

(2 marks)

##

(a) Any ***two*** sets of the following: (4)

- It is supported by C-shaped cartilage (1)
  which can keep the trachea open all the time. (1)
- Its inner lining contains mucus-secreting cells which produces mucus. (1)
  The mucus can trap the dust particles and bacteria. (1)
- Its inner lining is covered with cilia. (1)
  The cilia can remove the mucus trapped with dust particles and bacteria away from the respiratory tract. (1)

(b) Structure W is highly folded (1)
to increase the surface area for diffusion of gases. (1)
The wall of structure W is thin / one-cell thick (1)
to shorten the distance for the diffusion of gases. (1)
(Or other acceptable answers)

(c) When the man inhales, his diaphragm muscle contracts and (1)
the diaphragm / structure Z becomes flattened. (1)

(d) Air leaks into the pleural cavity. (1)
The lungs collapse because of their own elasticity. (1)

(12 marks)

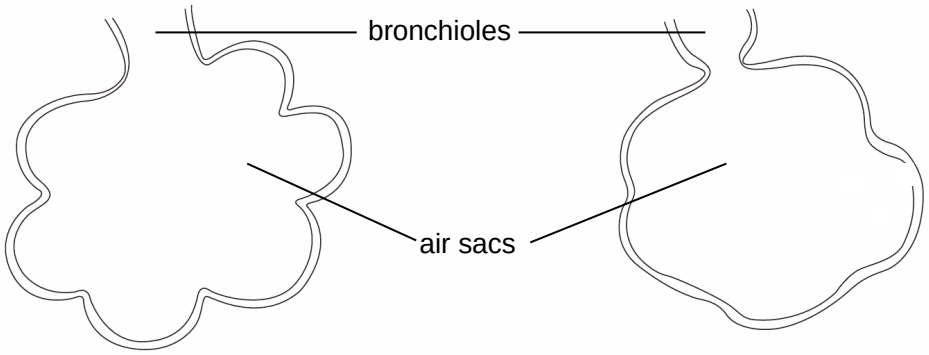
##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	2
(d)	Applying concepts	2

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|!|BCELB0080000019|!| ★

The diagrams below show the bronchioles and air sacs of two persons.



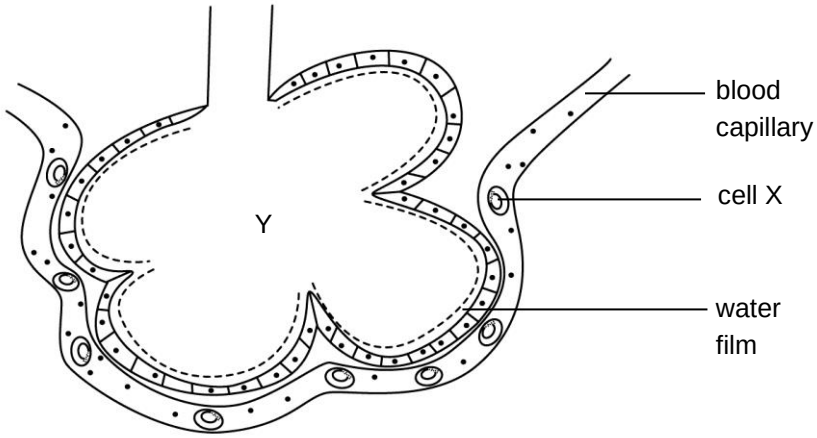
Person A **Person B**

(a) Which of the two persons, A or B, is most probably a smoker? (1 mark)

(b) With reference to the diagram, what are the ***two*** structural differences between the lungs of person A and person B? (2 marks)

(c) How would the differences in (b) affect the functioning of the lungs? (2 marks)

(d) The diagram below shows an air sac of person A as observed under a high-power microscope.



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(i) With reference to the diagram, state ***three*** adaptations of the air sac to gas exchange? (3 marks)

(ii) Write the route taken by oxygen to pass from space Y into cell X. (1 mark)

##

(a) Person B (1)

(b)

Person A	***Person B***	
- The air sac is more folded.	- The air sac is less folded.	(1)
- The diameter of the bronchiole is larger.	- The diameter of the bronchiole is smaller.	(1)

(c) For person B, the surface area for gas exchange decreases as the air sacs are less folded. (1)

In addition, person B has difficulty in breathing in or out due to the narrower passage of air at the bronchioles. (1)

(d) (i) The wall of the air sacs is one-cell thick to shorten the distance for the diffusion of gases. (1)

The lining is moist to dissolve the gases for diffusion. (1)

The air sac is surrounded by capillaries. This allows a rapid transport of gases to and away from the air sacs and maintains a steep concentration gradient for gas diffusion. (1)

(ii) Water film on the wall of the air sac → wall of the air sac → wall of the blood capillary → cell X (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Applying concepts	2
(d)(i)	Analysing data	3
(d)(ii)	Understanding basic concepts	1

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